



Coin Example: Analytical Solution

- The **posterior** is proportional to **likelihood** $\times$ **prior**

$$\Pr(\theta|y) \propto \Pr(y|\theta)\Pr(\theta)$$

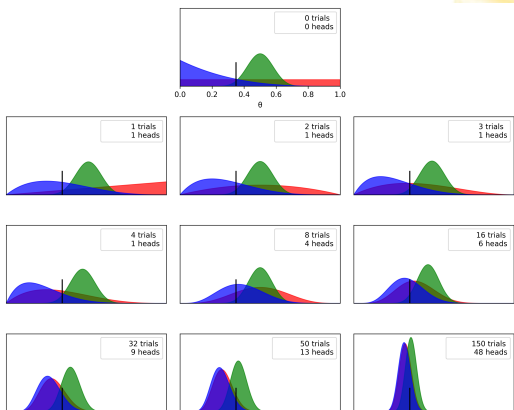
- Substituting **likelihood** with a Binomial and **prior** with a Beta

$$\begin{aligned}\Pr(\theta | Y) &= \underbrace{\frac{N!}{y!(N-y)!} \theta^y (1-\theta)^{N-y}}_{\text{likelihood}} \underbrace{\frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} \theta^{\alpha-1} (1-\theta)^{\beta-1}}_{\text{prior}} \\ &\propto \underbrace{\theta^y (1-\theta)^{N-y}}_{\text{likelihood}} \underbrace{\theta^{\alpha-1} (1-\theta)^{\beta-1}}_{\text{prior}} \\ &= \theta^{y+\alpha-1} (1-\theta)^{N-y+\beta-1} \\ &= \text{Beta}(\alpha_{\text{prior}} + y, \beta_{\text{prior}} + N - y)\end{aligned}$$

- This is how **the posterior is updated** given the data

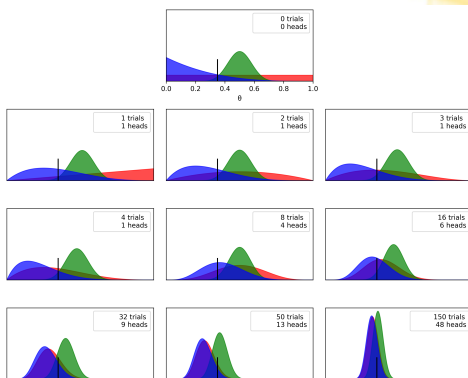
Coin Example: Effect of Priors (1/2)

- The true (unknown) value of the coin bias is 0.35
- Start with 3 different priors and update the model
 - **Red:** uniform prior
 - All bias values equally probable
 - **Green:** Gaussian-like prior around 0.5
 - Coin mostly unbiased
 - **Blue:** skewed towards tail
 - Coin biased
- Apply data to update the posterior distribution
- Update model



Coin Example: Effect of Priors (2/2)

- **Outcome of Bayesian analysis**
 - Posterior distribution, not a single value
- **Spread of posterior**
 - Proportional to uncertainty
 - Decreases with more data
 - Decreases faster if aligned with prior
 - With enough data, models with different priors converge to same result
- Applying posterior sequentially or at once yields same result



- *Frequentist vs Bayesian*
- Probabilistic Programming

Frequentist Approach vs Priors

- **Detractors of Bayesian approach** complain that:
 - *"One should let the data speak"*
 - The prior doesn't let the data speak for itself
-  **Counterpoints**
 - *"Data doesn't speak, but murmurs"*
 - Data doesn't have meaning per-se
 - Make sense of data only in context of models (e.g., mental models, mathematical models)
 - A prior is a mathematical model
 - Every statistical model has a prior, even if not explicit
 - Frequentist statistics still makes assumptions (i.e., has a prior), but are hidden
 - E.g., maximum likelihood estimate (MLE) in frequentist approach corresponds to a uniform prior and mode of the posterior
 - E.g., MLE is a point-estimate, not a distribution of plausible values

Advantages of Using Prior

- **Assumptions are clear and explicit**
 - Instead of hidden by frequentist or hacker ML approach
- **Prior**
 - Encourages deeper analysis of problem and data
 - Forces understanding before seeing data
- Posterior averaged over priors is **less prone to overfitting**
- Spread of distribution measures **uncertainty**
- Well-chosen prior simplifies and **speeds up inference**
 - *“When you encounter computational problems, there’s often an issue with your model”* (Gelman, 2008)

How to Choose Priors

- **Weakly-informative priors** (aka “flat”, “vague”, “diffuse priors”)
 - Provide minimal information
 - Coefficient of linear regression centered around 0: $\beta \sim \text{Normal}(0, 10)$
- **Regularizing priors**
 - Known information about the parameter
 - Parameter is positive: $\sigma \sim \text{HalfCauchy}(0, 5)$
 - Parameter close to zero, above/below a number, or in a range
 - $\beta \sim \text{Laplace}(0, 1)$ (lasso prior) encourages sparsity
 - $\beta \sim \text{Normal}(0, 1)$ discourages extreme values
- **Informative priors**
 - Strong priors from previous knowledge (expert opinion, studies)
 - From experimental data: $\beta_1 \sim \text{Normal}(2.5, 0.5^2)$
 - From previous data, about 5% of cases positive: $p \sim \text{Beta}(2, 38)$
- **Prior elicitation**
 - Compute least informative distribution given constraints
 - Estimate distribution using maximum entropy to satisfy constraints
 - E.g., beta distribution with 90% of mass between 0.1 and 0.7

Communicating the Model of a Bayesian Analysis

1. Communicate assumptions / hypothesis

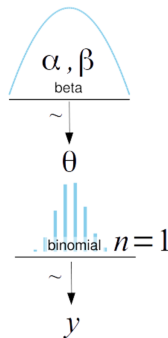
- Describe priors and probabilistic models
- E.g., coin-flip distributions:

$$\begin{cases} \theta \sim \text{Beta}(\alpha, \beta) \\ y \sim \text{Binomial}(n = 1, p = \theta) \end{cases}$$

2. Communicate Bayesian analysis result

- Describe posterior distribution
- Summarize location and dispersion
- Mean (or mode, median)
- Std dev
 - Misleading for skewed distributions
- Highest-posterior density (HPD)
 - Shortest interval containing a portion of probability density (e.g., 95% or 50%)
 - Amount is arbitrary (e.g., ArviZ defaults to 94%)

Kruschke diagram



Confidence Intervals vs Credible Intervals

- People confuse:
 - **Frequentist confidence intervals**
 - **Bayesian credible intervals**
- In the frequentist framework, there is a true (unknown) parameter value
 - A **confidence interval** may or may not contain the true parameter value
 - Interpretation of a 95% confidence interval
 - ❌ No: *"There is a 95% probability that the true value is in this interval"*
 - ✅ Yes: *"If repeated many times, 95% of intervals would contain the true value"*
- In the Bayesian framework, parameters are random variables
 - Interpretation of a 95% **Bayesian credible interval**
 - *"There is a 95% probability that the true parameter lies within this interval, given the observed data"*
 - Bayesian **credible interval** is intuitive

Confidence Intervals vs Credible Intervals (ELI5)

- **Confidence Interval (Frequentist)**

- Imagine fishing in a lake without seeing the fish
- You throw your net
- 95% confidence interval: *"If I threw this net 100 times, about 95 nets would catch the fish."*
- Important: Once the net is thrown, it either caught the fish or not. The 95% makes sense across many attempts

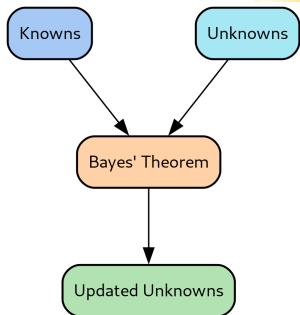
- **Credible Interval (Bayesian)**

- Imagine a magical map showing where fish *probably* are, based on past observations
- 95% credible interval: *"Given my map, there's a 95% chance the fish is inside this part of the lake."*
- The fish's location is uncertain, and probability describes your belief

- Frequentist vs Bayesian
- *Probabilistic Programming*

Bayesian Statistics

- Given:
 - The **"knows"**
 - Model structure (modeled as a graph of probability distributions)
 - Data, observations (modeled as constants)
 - The **"unknowns"**
 - Model parameters (modeled as probability distributions)
- Use Bayes' theorem to:
 - Condition unknowns to knowns
 - Reduce the uncertainty about the unknowns
- **Problem**
 - Most probabilistic models are analytically intractable
- **Solution**
 - Probabilistic programming
 - Specify a probabilistic model using code
 - Solve models using numerical techniques



Probabilistic Programming Languages

- **Steps:**

1. Specify models using code
2. Numerical models solve inference problems without need of user to understand how
 - Universal inference engines
 - PyMC3: flexible Python library for probabilistic programming
 - Theano: library to define, optimize, evaluate mathematical expressions using tensors
 - ArviZ: library to interpret probabilistic model results

- **Pros:**

- Compute results without analytical closed form
- Treat model solving as a black box
- Focus on model design, evaluation, interpretation

- **Probabilistic programming languages**

- Similar impact as Fortran on scientific computing
- Build algorithms but ignore computational details

Coin Example: Numerical Solution (1/3)

- It's a synthetic example!
 - Assume you know the true value of θ (not true in general)
- **Workflow**
 - Model the prior θ and the likelihood $Y|\theta$

$$\begin{cases} \theta \sim \text{Beta}(\alpha = 1, \beta = 1) \\ Y \sim \text{Binomial}(n = 1, p = \theta) \end{cases}$$

- Observe samples of the variable Y
- Run inference
- Generate samples of the posterior
- Summarize posterior
 - E.g., Highest-Posterior Density (HPD)
- ...

Coin Example: Numerical Solution (2/3)

- Generate data from ground truth model
- Build PyMC model matching mathematical model
- PyMC uses NUTS sampler, computes 4 chains
- No trace diverges
- Kernel density estimation (KDE) for posterior
- Should be Beta
- Traces appear “noisy” and non-diverging (good)
- Numerical summary of posterior: mean, std dev, HDI
- $E[\hat{\theta}] \approx 0.324$
- $\Pr(\hat{\theta} \in [0.031, 0.653]) = 0.94$

```
[18]: np.random.seed(123)
n = 4
# Unknown value.
theta_real = 0.35

# Generate some observational data.
data = stats.bernoulli.rvs(p=theta_real, size=n)
data
```

```
[18]: array([1, 0, 0, 0])
```

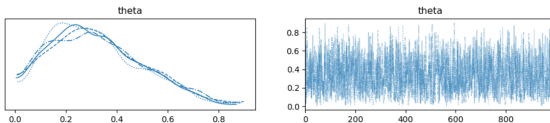
```
[19]: with pm.Model() as our_first_model:
    # Prior.
    theta = pm.Beta('theta', alpha=1., beta=1.)
    # Likelihood.
    y = pm.Bernoulli('y', p=theta, observed=data)
    # (Numerical) Inference to estimate the posterior distribution through samples.
    idata = pm.sample(1000, random_seed=123)
```

```
Auto-assigning NUTS sampler...
Initializing NUTS using jitter+adapt_diag...
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [theta]
```

Sampling 4 chains, 0 divergences ————— 100% 0:00:00 / 0:00:00

Sampling 4 chains for 1_000 tune and 1_000 draw iterations (4_000 + 4_000 draws total) took 1 seconds.

```
[20]: az.plot_trace(idata);
```



```
[21]: az.summary(idata)
```

	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	ess_bulk	ess_tail	r_hat
theta	0.324	0.179	0.031	0.653	0.005	0.003	1500.0	1737.0	1.0